I want you to create a playable browser game called **The Calibration Witness**. It should be a story-driven educational spy thriller where the player controls Mara Vale, a CIA scientist operating under cover as an international detector-calibration expert inside a fictional closed state called Karsovia. The game should teach real nuclear physics and quantum physics while also feeling like an espionage mystery. The balance should be roughly 50% science questions and 50% spycraft or field judgment. Do not make it feel like a classroom quiz pasted onto a spy story. The science should appear naturally through conversations, inspections, lab tours, technical disputes, and high-pressure professional situations where Mara must prove that she actually understands the physics.

Build this as a polished single-page web game in HTML, CSS, and JavaScript, with no external dependencies unless absolutely necessary. It should run locally by opening the HTML file. The UI should feel like a serious narrative game: main story panel, side panel with Mara's stats, email archive, notebook, collected clues, and day progression. Do not include any meta language in the game like "MVP," "prototype," "70% education," "oral exam," "hidden truth," "this is a game mechanic," or anything that breaks the story. Everything visible to the player should feel like part of the fiction.

The core mystery should be coherent from the beginning. Before writing scenes, define the actual truth of the case. For this version, the real situation is: Karsovia is hiding a mixed technical scandal. First, there was a medical-isotope contamination incident that officials concealed by rewriting labels, moving waste, and blaming detector calibration. Second, the same cover-up is being used to hide undeclared neutron-conditioning work connected to a quantum navigation or quantum sensing package. The quantum program is not a nuclear weapon, but it is strategically sensitive. The final truth is not "everything is evil" and not "nothing is wrong." It is a mixed case: a safety cover-up plus undeclared dual-use research.

The game should take place over eight days. Each day should have a clear beginning and an explicit end-of-day transition. Every day should include: 1. Mara's daily CIA email, 2. at least one scene with atmosphere and room signs, 3. one collectible case clue, 4. one or more science or field decisions, and 5. an end-of-day summary. The player should always have access to all past CIA emails through the side panel. Reading email should never be a main-scene decision. It is not something Mara chooses instead of acting. Mara expects one email from Langley every morning. It is the only communication channel the CIA can safely use while she is in-country. The message is disguised as a normal science-history circular in case anyone inspects her device, but Mara knows it is from the CIA. The article should not directly mention the mission. It should look like a long Wikipedia-style entry about one scientist, with interesting biography, history, discoveries, controversies, and scientific context. Hidden inside that ordinary educational material are concepts that will help Mara later. Questions should never say "based on today's article" or "using the email." The player should simply benefit from having read and understood it.

Use exactly one scientist per day, not pairs. Good choices are: Day 1 Henri Becquerel, Day 2 Marie Curie, Day 3 Ernest Rutherford, Day 4 Frederick Soddy, Day 5 James Chadwick, Day 6 Enrico Fermi, Day 7 Otto Stern, Day 8 Isidor Rabi. Each article should be substantial, at least 500 to 800 words if possible, and should feel encyclopedic: birth/death, context, key discovery, scientific importance, historical consequences, common misconceptions, and why the concept still matters. These articles should teach things like radioactivity as an atomic property, alpha/beta/gamma differences, isotopes, half-life, scattering and the nucleus, neutrons, moderation, fission, spin, resonance, and precision measurement. Again, the articles should never explicitly say "this will help Mara today."

Separate three types of information clearly. **CIA emails** are always available background knowledge. **Room signs and graphics** are atmospheric and help set the scene, but they should not give away the answer. **Case clues** are collectible evidence that proves what is actually happening. The room graphics should be cartoon-like and interesting, but they should not explain how to solve the physics problem. For example, a reactor diagram can show rods, tanks, pipes, and warning labels, but it should not label "moderator slows neutrons" or "control rods absorb neutrons." A gamma spectrum image can show peaks, axes, and handwritten marks, but not explain what peaks mean. The explanations should live in the CIA articles, dialogue, feedback after answers, or the player's own reasoning.

There should be exactly one major collectible clue per day. The clues should form a logical proof chain. Each clue should either establish a fact, connect two locations, weaken a false explanation, or turn a suspicious anomaly into a defensible scientific conclusion. The player should be able to open a "Case Clues" folder and review each collected clue. The final accusation should depend on how many clues the player collected and whether they interpreted them correctly.

Use this clue chain:

Day 1 clue: A customs dosimeter offset log. It shows several border detectors were recently recalibrated after an unexplained above-background reading. This does not prove wrongdoing, but it establishes that officials already knew about anomalous radiation.

Day 2 clue: A medical isotope shipping label with a date mismatch. It suggests the official processing date does not match the activity history.

Day 3 clue: A waste barrel activity log. The activity has dropped by a factor of two over eight days, contradicting the newer label. This points to label rewriting and teaches half-life reasoning.

Day 4 clue: A gamma spectrum printout with an unlabeled peak. The peak is not explained by ordinary background or a simple calibration failure. This weakens the "bad detector only" explanation.

Day 5 clue: A badge-dose ledger for a technician who entered a supposedly clean corridor. This connects the isotope center to a human safety incident and shows why the cover-up matters.

Day 6 clue: An activated metal coupon report. The sample was stable before neutron exposure and later emitted gamma rays. This introduces neutron activation and suggests neutron work beyond routine medical isotope handling.

Day 7 clue: A maintenance order for Hall 9 shielding and vibration isolation. It connects neutron activation evidence to a restricted quantum or inertial-sensing laboratory.

Day 8 clue: A timing-module transport manifest. It links Hall 9's quantum package to the neutron-conditioned materials and shows the program is strategic dual-use, not just a hospital accident.

The final conclusion should only be defensible if the player has the chain: anomalous radiation, rewritten isotope dates, half-life contradiction, unexplained gamma peak, technician exposure, neutron activation, Hall 9 infrastructure, and timing module transport. A good ending should say Mara does not overclaim nuclear weapons work. She reports a concealed contamination incident plus undeclared neutron-conditioning work for a quantum navigation/sensing system. Bad endings include: she overclaims a weapons program, she dismisses everything as calibration error, she solves the physics but breaks cover, or she collects clues but cannot connect them.

Science questions should be real fundamentals of nuclear and quantum physics. Examples: What is the difference between dose and dose rate? Why is alpha radiation more dangerous inside the body than outside? If U has Z = 92, how many neutrons are in U-235? What does one half-life mean for activity? Why do gamma spectra have discrete peaks? What distinguishes contamination from activation? Why do neutrons matter if they have no charge? Why can slowing neutrons increase fission probability in U-235? What do control rods do versus moderators versus coolant? What did Stern-Gerlach show about quantized spin? What is resonance in a quantum system? Why can interferometers detect tiny path changes? What is tunneling in an STM? Why does quantum key distribution reveal some eavesdropping but not secure the endpoints?

Spycraft decisions should be real field judgment, not arbitrary stealth choices. Examples: when to challenge a lie directly versus ask for a repeat measurement; whether to protect a frightened technician; whether to preserve a copied spectrum or return it to avoid suspicion; whether to use a reception conversation to test a scientist’s story; whether to report a partial truth or keep gathering evidence; whether to let a host believe she thinks the issue is calibration while she investigates deeper. Good spycraft should be tied to science. Mara’s cover works because she behaves like a careful scientist: precise, skeptical, calm, and reluctant to overclaim.

Vary the correct answer positions. Do not make the best answer usually A. Shuffle choices deliberately so correct answers appear in A, B, C, and D across the game. Also include partial answers. The feedback should explain why an answer is expert, partial, wrong, or dangerous. Wrong science should not always end the game. It can trigger suspicion, lower trust, or force a remedial scene. Wrong spycraft can close a social route but open a worker route. The game should fail forward.

Create a polished narrative tone. Mara should feel like a real person under pressure, not a quiz avatar. Give her small observations: cold train windows, institutional lighting, overwatered plants in ministry halls, a technician’s shaking hand, old Soviet-style physics murals, the smell of ozone near a laser table, the sound of ventilation in a sealed corridor. NPCs should have personalities: a proud nuclear physicist, a frightened technician, a smooth ministry official, a precise quantum metrologist, a reactor engineer who respects competence, a security officer who watches for contradictions. The story should become more tense as the clues accumulate.

Deliver a complete playable HTML file. Include CSS styling, JavaScript game state, scene data, email archive, clue collection, notebook, stat tracking, day transitions, and multiple endings. The game should be understandable and playable immediately. The code should be clean and easy to expand.